Tak Yeon Lee

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Education

University of Maryland — *College Park, MD, USA*

Ph.D Computer Science — *2009.1 - 2017.5*

- Thesis: Toward Symbiotic Human-AI Interaction Focusing on Programming by Example
- Advisor: Prof. Benjamin B. Bederson

Delft University of Technology — *Delft, Netherlands*

MSc. Industrial Design Engineering — *2004.9 - 2006.11*

- Advisor: Prof. Pieter Jan Stappers

Korea Advanced Institute of Science and Technology — *Daejeon, Republic of Korea*

BSc. Industrial Design — *1996.3 - 2004.2*

- Minors in Computer Science

Professional Experience

KAIST — *Daejeon, Korea*

Associate Professor of Industrial Design — *2021.3 - CURRENT*

- Director of AEL (https://ai-experience-lab.github.io/)
- Affiliate Faculty: College of AI

Adobe Research — *San Jose, USA*

Research Scientist — *2017.5 - 2021.2*

- Data science and visualization technology for Adobe Experience Cloud

IBM Research — *Cambridge, MA, USA*

Research Intern — *2012.6 - 2012.8*

- Published "Experiments on motivational feedback for crowdsourced workers" (ICWSM 2013)
- Mentor: Casey Dugan and Werner Geyer

Microsoft — *Redmond, WA, USA*

UX Design Intern — *2010.6 - 2010.8*

Publications

Journals

1 J. Park, E. Park, D. Lee, S. Kang, **Takyeon Lee**, H. Hong, S.-J. Lee, H. Kim, Y. R. Park, and U. Lee, "Human digital twins for pervasive healthcare: A scoping review," *Health Informatics Journal*, vol. 31, no. 4, p. 14 604 582 251 392 455, 2025, PMID: 41166748. DOI: 10.1177/14604582251392455.

*2 Y. S. Kim, H. S. Moon, S. Lee, and **Lee, Tak Yeon**, "'histochat': Leveraging ai-driven historical personas for personalized and engaging middle school history education," *Proc. ACM Hum.-Comput. Interact.*, vol. 9, no. 7, Oct. 2025. DOI: 10.1145/3757534.

*3 **Tak Yeon Lee**, A. Smith, K. Seppi, N. Elmqvist, J. Boyd-Graber, and L. Findlater, "The human touch: How non-expert users perceive, interpret, and fix topic models," *International Journal of Human-Computer Studies*, vol. 105, pp. 28–42, 2017, ISSN: 1071-5819. DOI: https://doi.org/10.1016/j.ijhcs.2017.03.007.

*: Published at venues that are qualifed as S-class or Top Conferences at KAIST

†: Co-corresponding author

Conference Proceedings

*1 Y. Jung, T. Hoang Anh Vo, H. S. Moon, J. Y. Choi, H. Oh, U. Lee, E. Kim, **Lee, Tak Yeon**†, and U. Lee, "Toward Flexible Psychiatric History-Taking and Visualization: Exploring Clinician Perspectives with Large Language Models," en, in *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, Barcelona Spain: ACM, Apr. 2026, pp. 1–23, ISBN: 979-8-4007-2278-3. DOI: 10.1145/3772318.3790970.

*2 S. Park, **Lee, Tak Yeon**†, and W. Lee, "LLM-box vs. Thinking-box: Designing for Deliberate User Engagement with Distorted Information in Conversational Search," en, in *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, Barcelona Spain: ACM, Apr. 2026, pp. 1–20, ISBN: 979-8-4007-2278-3. DOI: 10.1145/3772318.3790271.

3 H. Lee, Y. Oh, H. S. Moon, and **Lee, Tak Yeon**, "Understanding How Creators Collaborate with Conversational Agents in Short-Form Video Production," en, in *Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems*, Barcelona , Spain: ACM, Apr. 2026, pp. 1–6, ISBN: 979-8-4007-2281-3. DOI: 10.1145/3772363.3798927.

*4 S. G. Kim, J. Y. Choi, P. Y. Lee, J. Chung, R. Rossi, J. Kil, E. Koh, and **Lee, Tak Yeon**, "Grounded generation of embellished bar chart ensuring chart integrity," in *2025 IEEE Visualization and Visual Analytics (VIS)*, 2025, pp. 101–105. DOI: 10.1109/VIS60296.2025.00026.

5 J. Y. Choi, S. G. Kim, J. Jeong, R. A. Rossi, J. Kil, and **Lee, Tak Yeon**, "Gaze2prompt: Turning eye-tracking data into visual prompts for multimodal llms," in *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing*, ser. UbiComp Companion '25, Finland: Association for Computing Machinery, 2026, pp. 110–114, ISBN: 9798400714771. DOI: 10.1145/3714394.3754401.

*6 R. Luera, R. Rossi, F. Dernoncourt, A. Siu, S. Kim, T. Yu, R. Zhang, X. Chen, N. Lipka, Z. Zhang, S. Gyeom Kim, and T. Yeon Lee, "Personalizing data delivery: Investigating user characteristics and enhancing llm predictions," in *Companion Proceedings of the ACM on Web Conference 2025*, ser. WWW '25, Sydney NSW, Australia: Association for Computing Machinery, 2025, pp. 1167–1171, ISBN: 9798400713316. DOI: 10.1145/3701716.3715452.

7 J. Chung, S. G. Kim, and **Lee, Tak Yeon**, "Wrighthere: Supporting children's creative writing with ai-infused interactive 3d environment," in *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, ser. CHI EA '25, Association for Computing Machinery, 2025, ISBN: 9798400713958. DOI: 10.1145/3706599.3719932.

8 J. Jeong and **Lee, Tak Yeon**, "Ligs: Developing an llm-infused game system for emergent narrative," in *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, ser. CHI EA '25, Association for Computing Machinery, 2025, ISBN: 9798400713958. DOI: 10.1145/3706599.3720212.

9 H. Bae, N. Kim, S. Lee, and **Lee, Tak Yeon**, "Bridging bond beyond life: Designing vr memorial space with stakeholder collaboration via research through design," in *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, ser. CHI EA '25, Association for Computing Machinery, 2025, ISBN: 9798400713958. DOI: 10.1145/3706599.3719871.

10 S. G. Kim, J. Y. Choi, R. Rossi, E. Koh, and **Lee, Tak Yeon**, "Chart-to-experience: Benchmarking multimodal llms for predicting experiential impact of charts," in *2025 IEEE 18th Pacific Visualization Conference (PacificVis)*, 2025, pp. 340–345. DOI: 10.1109/PacificVis64226.2025.00040.

11 H. Im, T. Kim, E. Jung, B. Ku, S. Baek, Y.-k. Lim, T.-J. Nam, and **Lee, Tak Yeon**, "Virfie: Enhancing remote togetherness with user-created scenarios for virtual group selfie," in *Intelligent Systems and Applications*, K. Arai, Ed., Cham: Springer Nature Switzerland, 2024, pp. 628–645, ISBN: 978-3-031-66336-9.

*12 J. Han, H. Yoo, J. Myung, M. Kim, **Lee, Tak Yeon**, S.-Y. Ahn, and A. Oh, "RECIPE4U: Student-ChatGPT interaction dataset in EFL writing education," in *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, N. Calzolari, M.-Y. Kan, V. Hoste, A. Lenci, S. Sakti, and N. Xue, Eds., Torino, Italia: ELRA and ICCL, May 2024, pp. 13 666–13 676.

13 C. Harris, R. A. Rossi, S. Malik, J. Hoffswell, F. Du, **Lee, Tak Yeon**, E. Koh, and H. Zhao, "Visual insight recommendation: From ranking insight visualizations to insight types," in *2023 IEEE International Conference on Big Data (BigData)*, 2023, pp. 1716–1725. DOI: 10.1109/BigData59044.2023.10386099.

14 J. Han, H. Yoo, Y. Kim, J. Myung, M. Kim, H. Lim, J. Kim, **Lee, Tak Yeon**, H. Hong, S.-Y. Ahn, and A. Oh, "Recipe: How to integrate chatgpt into efl writing education," in *Proceedings of the Tenth ACM Conference on Learning Scale*, ser. LS '23, Copenhagen, Denmark: Association for Computing Machinery, 2023, pp. 416–420, ISBN: 9798400700255. DOI: 10.1145/3573051.3596200.

*15 C. Harris, R. Rossi, S. Malik, J. Hoffswell, F. Du, **Lee, Tak Yeon**, E. Koh, and H. Zhao, "Spotlight: Visual insight recommendation," in *Companion Proceedings of the ACM Web Conference 2023*, ser. WWW '23 Companion, Austin, TX, USA: Association for Computing Machinery, 2023, pp. 19–23, ISBN: 9781450394192. DOI: 10.1145/3543873.3587302.

*16 X. Qian, R. A. Rossi, F. Du, S. Kim, E. Koh, S. Malik, **Lee, Tak Yeon**, and N. K. Ahmed, "Personalized visualization recommendation," 3, vol. 16, New York, NY, USA: Association for Computing Machinery, Sep. 2022. DOI: 10.1145/3538703.

17 H. Im, T. Kim, E. Jung, B. Ku, S. Baek, and **Lee, Tak Yeon**, "Virfie: Virtual group selfie station for remote togetherness," in *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems*, ser. CHI EA '22, New Orleans, LA, USA: Association for Computing Machinery, 2022, ISBN: 9781450391566. DOI: 10.1145/3491101.3519767.

*18 Z. Zeng, P. Moh, F. Du, J. Hoffswell, **Lee, Tak Yeon**, S. Malik, E. Koh, and L. Battle, "An evaluation-focused framework for visualization recommendation algorithms," 1, vol. 28, 2022, pp. 346–356. DOI: 10.1109/TVCG.2021.3114814.

*19 X. Qian, R. A. Rossi, F. Du, S. Kim, E. Koh, S. Malik, **Lee, Tak Yeon**, and J. Chan, "Learning to recommend visualizations from data," in *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*, ser. KDD '21, Virtual Event, Singapore: Association for Computing Machinery, 2021, pp. 1359–1369, ISBN: 9781450383325. DOI: 10.1145/3447548.3467224.

*20 Y. Xian, H. Zhao, **Lee, Tak Yeon**, S. Kim, R. Rossi, Z. Fu, G. de Melo, and S. Muthukrishnan, "Exacta: Explainable column annotation," in *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*, ser. KDD '21, Virtual Event, Singapore: Association for Computing Machinery, 2021, pp. 3775–3785, ISBN: 9781450383325. DOI: 10.1145/3447548.3467211.

*21 X. Qian, E. Koh, F. Du, S. Kim, J. Chan, R. A. Rossi, S. Malik, and **Lee, Tak Yeon**, "Generating accurate caption units for figure captioning," in *Proceedings of the Web Conference 2021*, ser. WWW '21, Ljubljana, Slovenia: Association for Computing Machinery, 2021, pp. 2792–2804, ISBN: 9781450383127. DOI: 10.1145/3442381.3449923.

22 N. Lipka, **Lee, Tak Yeon**, and E. Koh, "Detecting link and landing page misalignment in marketing emails," in *Database and Expert Systems Applications*, M. Elloumi, M. Granitzer, A. Hameurlain, C. Seifert, B. Stein, A. M. Tjoa, and R. Wagner, Eds., Cham: Springer International Publishing, 2018, pp. 254–264, ISBN: 978-3-319-99133-7.

23 **Lee, Tak Yeon** and E. Koh, "Identifying types of misalignments between promotion emails and landing pages," in *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems*, ser. CHI EA '18, Montreal QC, Canada: Association for Computing Machinery, 2018, pp. 1–6, ISBN: 9781450356213. DOI: 10.1145/3170427.3188684.

24 **Lee, Tak Yeon**, C. Dugan, and B. B. Bederson, "Towards understanding human mistakes of programming by example: An online user study," in *Proceedings of the 22nd International Conference on Intelligent User Interfaces*, ser. IUI '17, Limassol, Cyprus: Association for Computing Machinery, 2017, pp. 257–261, ISBN: 9781450343480. DOI: 10.1145/3025171.3025203.

*25 A. Smith, **Lee, Tak Yeon**, F. Poursabzi-Sangdeh, J. Boyd-Graber, N. Elmqvist, and L. Findlater, "Evaluating visual representations for topic understanding and their effects on manually generated topic labels," L. Lee, M. Johnson, and K. Toutanova, Eds., vol. 5, Cambridge, MA: MIT Press, 2017, pp. 1–16. DOI: 10.1162/tacl_a_00042.

26 **Lee, Tak Yeon** and B. B. Bederson, "Give the people what they want: Studying end-user needs for enhancing the web," vol. 2, Nov. 2016, e91. DOI: 10.7717/peerj-cs.91.

27 **Tak Yeon Lee**, M. L. Mauriello, J. Ahn, and B. B. Bederson, "Ctarcade: Computational thinking with games in school age children," 1, vol. 2, 2014, pp. 26–33. DOI: https://doi.org/10.1016/j.ijcci.2014.06.003.

28 **Lee, Tak Yeon**, C. Dugan, W. Geyer, T. Ratchford, J. Rasmussen, N. S. Shami, and S. Lupushor, "Experiments on motivational feedback for crowdsourced workers," 1, vol. 7, Aug. 2013, pp. 341–350. DOI: 10.1609/icwsm.v7i1.14428.

29 **Lee, Tak Yeon**, M. L. Mauriello, J. Ingraham, A. Sopan, J. Ahn, and B. B. Bederson, "Ctarcade: Learning computational thinking while training virtual characters through game play," in *CHI '12 Extended Abstracts on Human Factors in Computing Systems*, ser. CHI EA '12, Austin, Texas, USA: Association for Computing Machinery, 2012, pp. 2309–2314, ISBN: 9781450310161. DOI: 10.1145/2212776.2223794.

30 M. Rios-Berrios, P. Sharma, **Tak Yeon Lee**, R. Schwartz, and B. Shneiderman, "Treecovery: Coordinated dual treemap visualization for exploring the recovery act," 2, vol. 29, 2012, pp. 212–222. DOI: https://doi.org/10.1016/j.giq.2011.07.004.

Awards

2026 **WrightHere**, iF Design Award

2025 **Chart-to-Experience**, Best VisNote, IEEE PacificVis

2024 **Dear.**, iF Design Award

Patents

Jan 2026 **Machine learning techniques for generating visualization recommendations**, Patent, US 19343670

Jun 2024 **Systems for generating interactive reports**, Patent, US 17474188

May 2024 **Lifelong schema matching**, Patent, US 17036453

Dec 2023 **Facilitating generation of data visualizations via natural language processing**, Patent, US 17354954

Oct 2023 **Configuration of user interface for intuitive selection of insight visualizations**, Patent, US 17161770

Oct 2023 **Generating visual data stories**, Patent, US 18069561

Oct 2023 **Automated caption generation from a dataset**, Patent, US 17094435

Aug 2023 **Personalized visualization recommendation system**, Patent, US 17091941

May 2023 **Generating explanatory paths for predicted column annotations**, Patent, US 16796681

Feb 2023 **Dynamic user interface and machine learning tools for generating digital content and multivariate testing recommendations**, Patent, US 17383114

Jan 2023 **Dynamically determining schema labels using a hybrid neural network encoder**, Patent, US 16751755

Oct 2022 **Detecting and resolving semantic misalignments between digital messages and external digital content**, Patent, US 15910926

May 2022 **Identifying and presenting misalignments between digital messages and external digital content**, Patent, US 16419676

Mar 2022 **Graph-based configuration of user interface for selection of features in visualization applications**, Patent, US 17015495

Oct 2021 **Systems for suggesting content components**, Patent, US 17183055

Invited Talks

2023.03 **How to Live in the Era of Generative AI**, KAIST

2024.11 **The Role of Designers in the Age of AI**, KAIST AI Executive Program

2024.12 **UX Designers in the Age of AI: Roles and Opportunities**, LG

2024.11 **Object-Oriented Design in the Age of AI**, Design 3.0 Forum

2024.11 **Large-scale Persona Generation and UX Testing with AI and Small Data**, Korea University

2024.07 **AI as a Design Material**, LG

2024.03 **AI Utilization Strategies for UX Designers**, Samsung SDS

2024.02 **AI Utilization Strategies for UX Designers**, INITION

2023.08 **AI and Visual Arts**, Ministry of Culture, Sports and Tourism

2023.07 **Understanding AI, Understanding Humans**, Moontak Network

2023.04 **How to Live in the Era of Generative AI**, Korean Intellectual Property Office

2022.10 **On Convergence: Global Conference**, Ringle

2022.04 **Designers in the Age of AI and Data**, Korean Society of Design Convergence

2021.11 **Design and Technology**, Korea Institute of Design Promotion

2021.03 **At the Intersection of Design and AI**, Handong Global University

2021.07 **Designing AI-infused Products for Smart Life**, Seoul National University Medical School

2021.03 **What if We Apply Data-Driven Decision Making to Product Design?**, LG Electronics

2021.05 **Toward Human-Centered AI for Healthcare Workers**, Akasa (USA)

Teaching Experience

Spring 2026 **Design Project 1 (ID.50003)** *KAIST*

Fall 2025 **Data Analytics for Designers (ID.40018)** *KAIST*

Spring 2025 **System Design (ID.40003)** *KAIST*

Spring 2025 **Design Special Topics V: Designing Artificial Human Behavior and Experience (ID.49902)** *KAIST*

Fall 2024 **Data Analytics for Designers (ID.40018)** *KAIST*

Spring 2024 **System Design (ID.40003)** *KAIST*

Fall 2023 **Design Special Topics V: Data Analytics for Designers (ID.49902)** *KAIST*

Spring 2023 **System Design (ID.40003)** *KAIST*

Fall 2022 **Design Special Topics V: Data Analytics for Designers (ID.49902)** *KAIST*

Fall 2022 **Freshman Seminar: School of Industrial Design (HSS.10090)** *KAIST*

Spring 2022 **System Design (ID.40003)** *KAIST*

Fall 2021 **Smart Life: AI-based Service Design (CoE.49001)** *KAIST*

Fall 2021 **Design Special Topics V: Data Analytics for Designers (ID.49902)** *KAIST*

Spring 2021 **Design Project (ID.50003)** *KAIST*

Service and Outreach

2026 – 2027	**PacificVis 2027**, Conference Sponsor Chair	*Busan, Korea*
2023	**“AI Design Park” Strategic Planning Initiative**, Lead Advisory Committee Member	*KEIT*
2024	**Design Innovation Technology Roadmap Initiative**, Advisory Committee Member	*MOTIE*
2022	**SIGCHI**, Local Event Moderator	*Seoul, Korea*
2021 – 2022	**TEI 2022**, Conference Local Chair	*Daejeon, Korea*